Curried Functional Spaces

Roman Maksimovich
Supervisor: JIN, Tianling

Hong Kong University of Science and Technology

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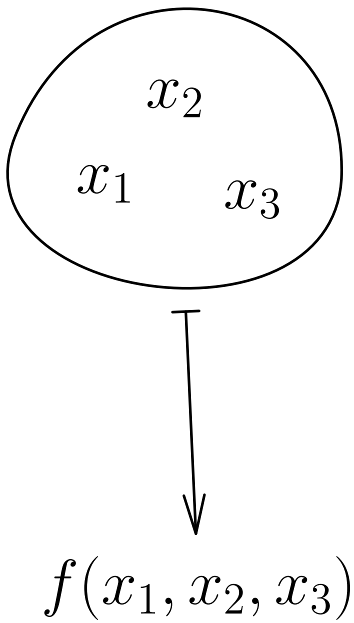
Premise

Multivariable functions

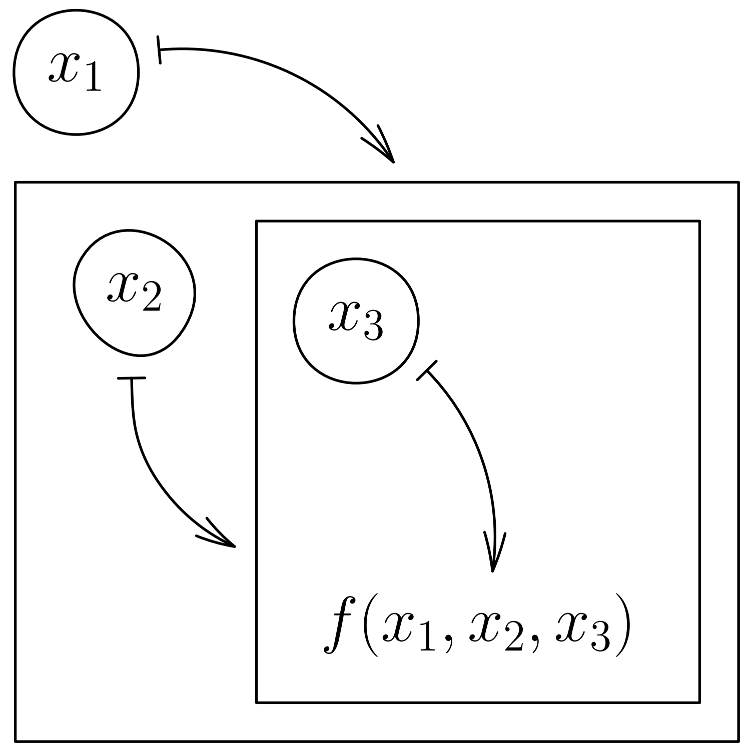
$$\mathbb{R}^n \to \mathbb{R}$$



Multivariate calculus

Curried functions

$$\mathbb{R} \to (\mathbb{R} \to (\ldots (\mathbb{R} \to \mathbb{R})))$$



λ-calculus,
functional programming

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Curried functions

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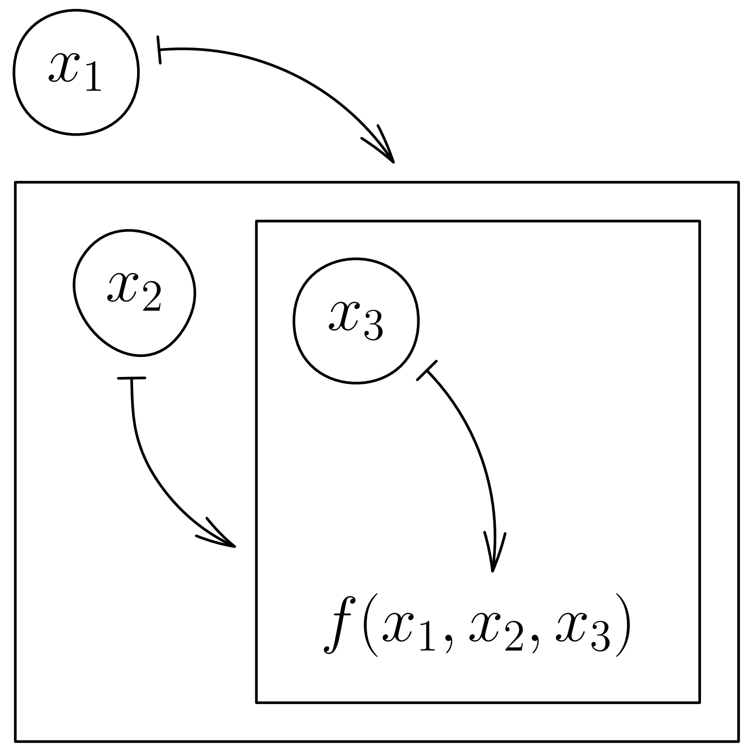

λ-calculus,
functional programming

?

Multivariate calculus

Continuous Functions

$C_n(\mathbb{R})$ — the space of continuous n-ary curried functions:

1. $C_0(\mathbb{R}) := \mathbb{R}$
2. $C_{n+1}(\mathbb{R}) := \{f \colon \mathbb{R} \to C_n(\mathbb{R}) \mid f \text{ is continuous}\}$

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what is the topology on $C_n(\mathbb{R})$?

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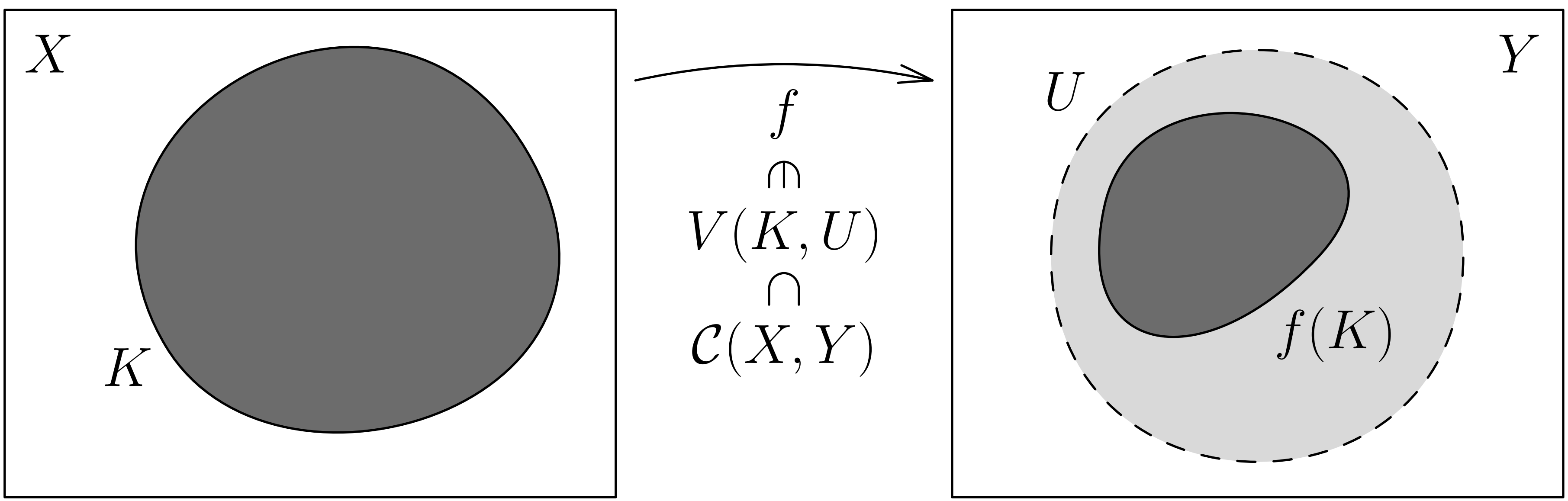
compact-open topology: $C_n(\mathbb{R}) = \mathcal{C}(\mathbb{R}, C_{n-1}(\mathbb{R}))$



Topological Vector Spaces

Remark: If $f_1, f_2 \in C_n(\mathbb{R})$, then $\alpha f_1 + \beta f_2 \in C_n(\mathbb{R})$.

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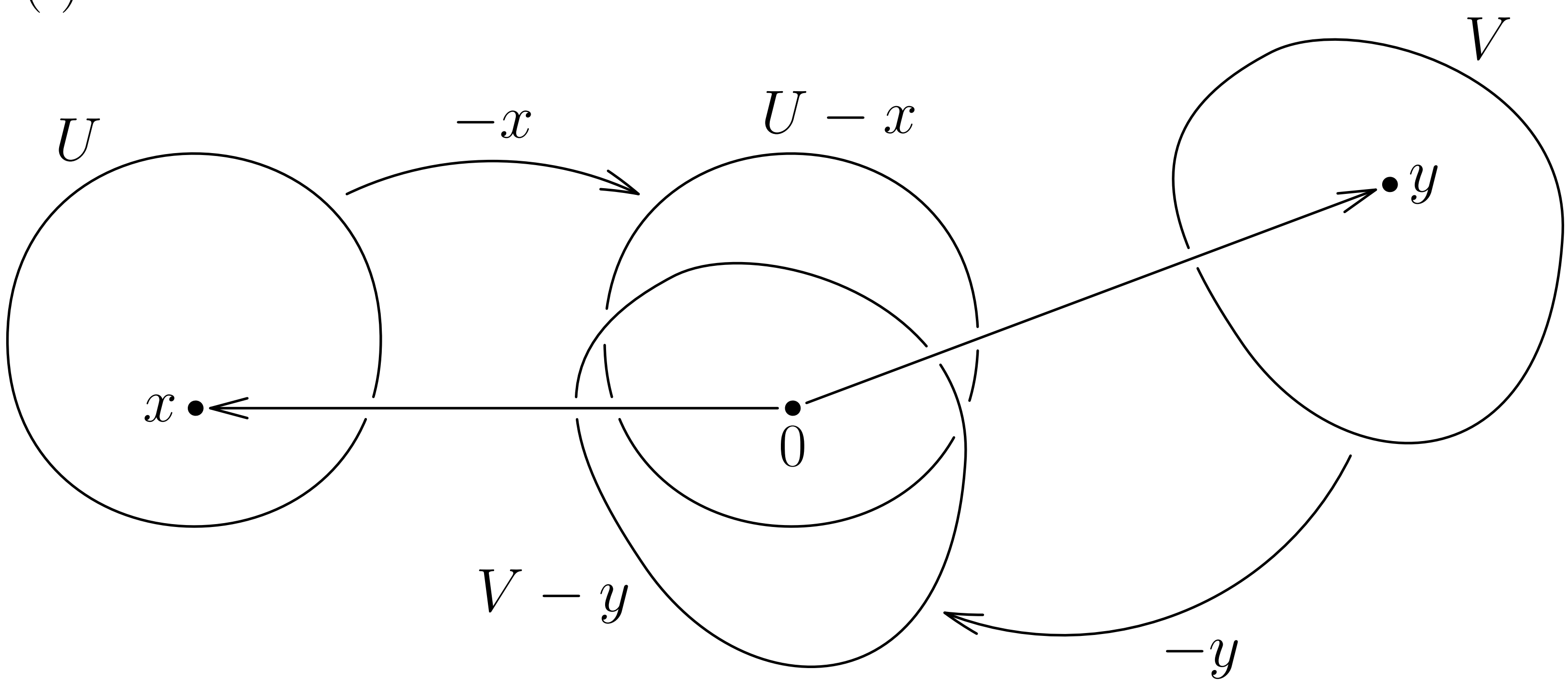
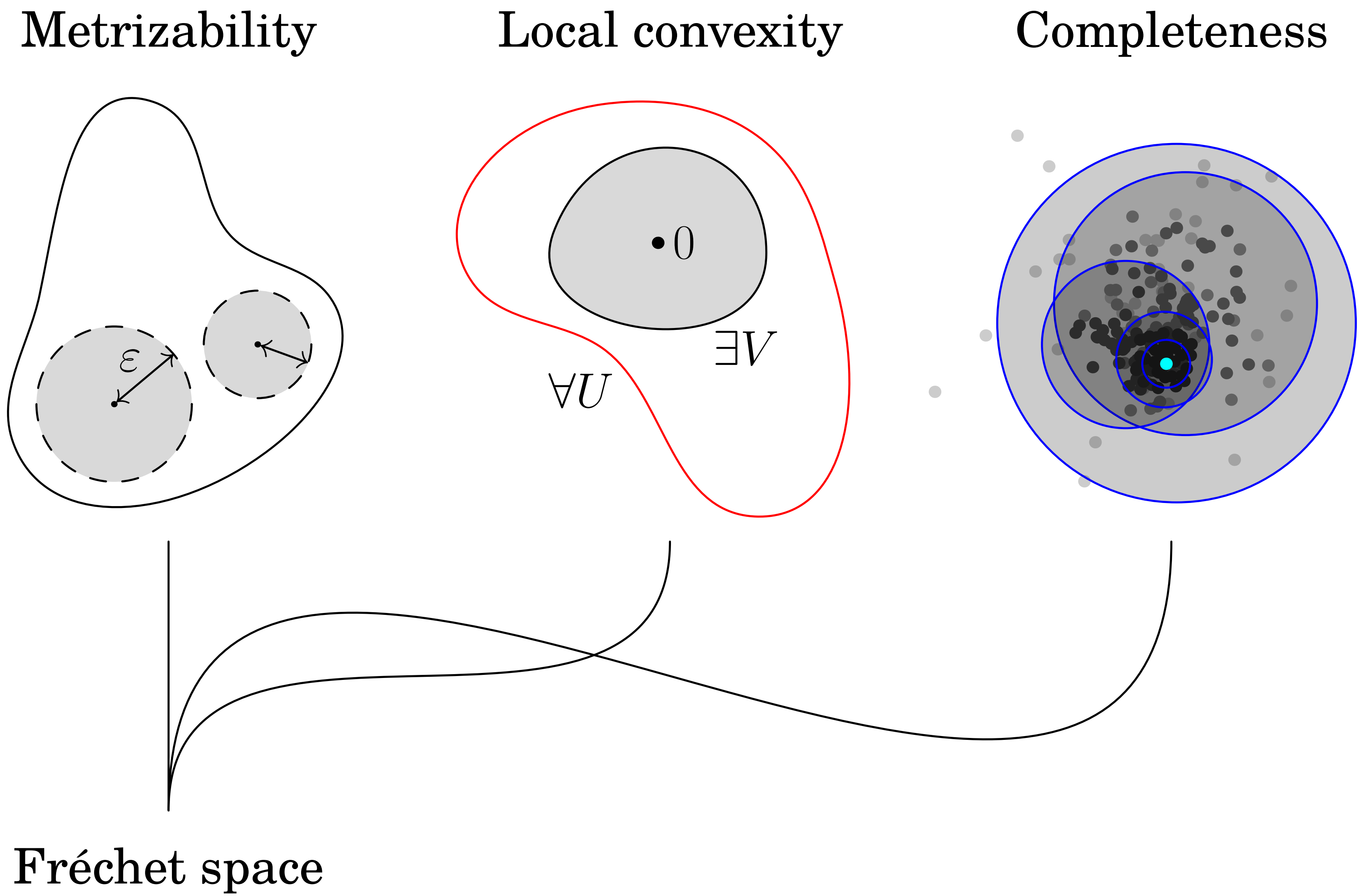
Definition: X is a TVS $\Leftrightarrow$ X has topological *and* vector structure, and they are *compatible*:

1. $(+)\colon X \times X \to X$ is continuous;
2. $(\cdot)\colon \mathbb{R} \times X \to X$ is continuous.



Properties of Topological Vector Spaces



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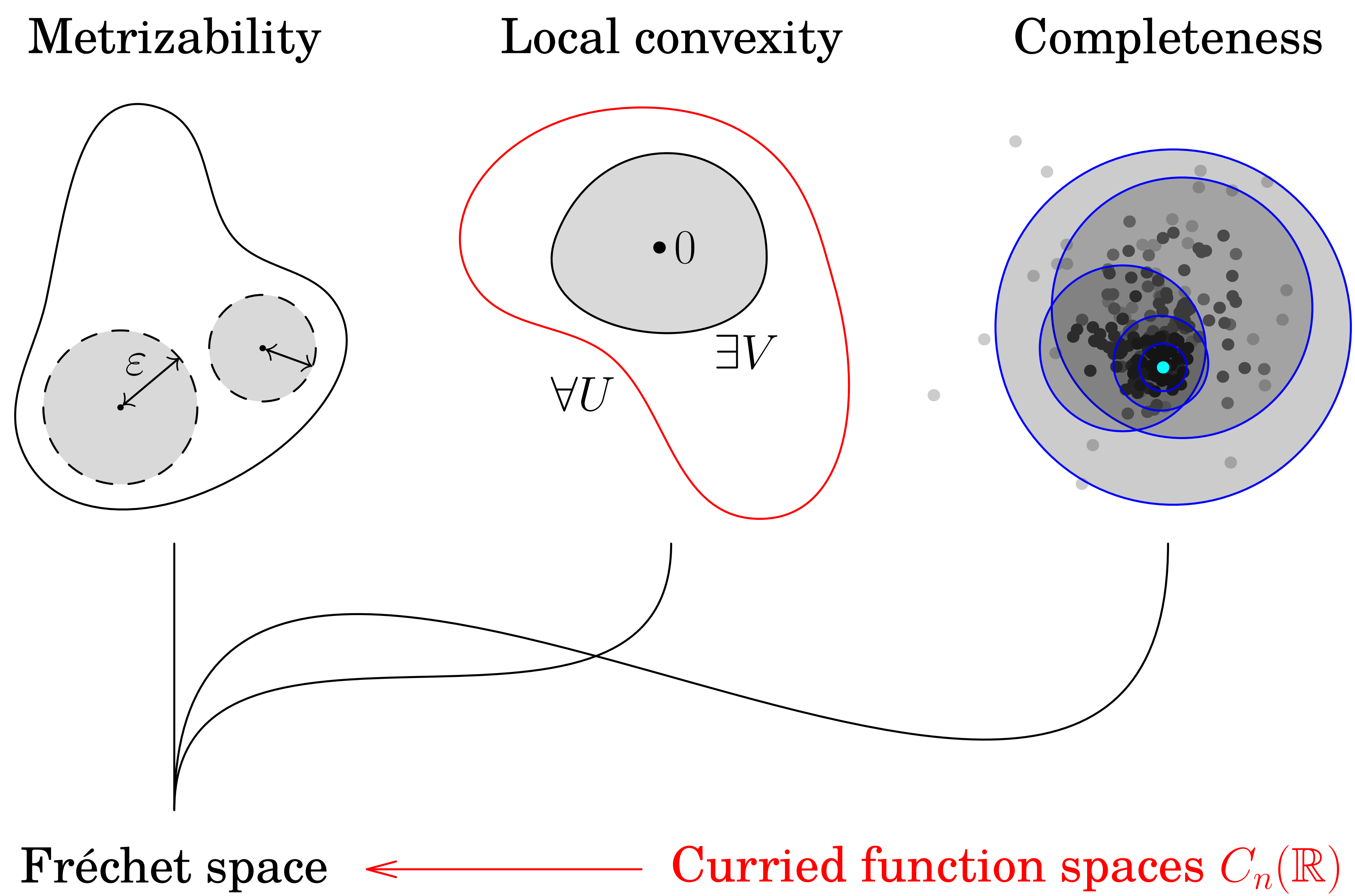

Continuous Functions

$C_n(\mathbb{R})$ — the space of continuous n-ary curried functions:

1. $C_0(\mathbb{R}) := \mathbb{R}$;
2. $C_{n+1}(\mathbb{R}) := \mathcal{C}(\mathbb{R}, C_n(\mathbb{R}))$, with the compact-open topology.

Results: for all $n \in \mathbb{N}$,

1. $C_n(\mathbb{R})$ is a Hausdorff topological vector space;
2. There is an isomorphism $\gamma\colon \mathcal{C}(\mathbb{R}^n, \mathbb{R}) \leftrightarrow C_n(\mathbb{R})$;
3. $C_n(\mathbb{R})$ is a Fréchet space.

$\Rightarrow$ Theorems from functional analysis apply:

1. Uniform boundedness principle;
2. Open mapping theorem;
3. $\ldots$

Differentiable functions

$D^m_n(\mathbb{R})$ — the space of n-ary m times differentiable functions:

1. $D^m_0(\mathbb{R}) := \mathbb{R}$;
2. $D^0_n(\mathbb{R}) := C_n(\mathbb{R})$;
3. If $n, m > 0$, then $D^m_n(\mathbb{R})$ consists of all such $f \in \mathcal{C}(\mathbb{R}, D^m_{n-1}(\mathbb{R}))$, that there is $f' \in D^{m-1}_n(\mathbb{R})$ with
$$\frac{f(x+h) - f(x)}{h} \xrightarrow[h \to 0]{} f'(x).$$

Differentiable functions

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pointwise convergence!

$$f_k \xrightarrow[k\to\infty]{} f \Leftrightarrow f_k(x_1)\dots(x_n) \xrightarrow[k\to\infty]{} f(x_1)\dots(x_n)$$

Differentiable functions

$D_n^m(\mathbb{R})$ — the space of n-ary m times differentiable functions:

1. $D_0^m(\mathbb{R}) := \mathbb{R}$;
2. $D_n^0(\mathbb{R}) := C_n(\mathbb{R})$;
3. If $n, m > 0$, then $D_n^m(\mathbb{R})$ consists of all such $f \in \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R}))$, that there is $f' \in D_n^{m-1}(\mathbb{R})$ with
$$\frac{f(x+h) - f(x)}{h} \xrightarrow[h \to 0]{} f'(x).$$

The topology on D_n^m is *coarsest* such that the maps

$$\begin{aligned} i\colon D_n^m(\mathbb{R}) &\to \mathcal{C}(\mathbb{R},\quad D_{n-1}^m(\mathbb{R})), \\ f &\mapsto f \end{aligned} \qquad\qquad \begin{aligned} d\colon D_n^m(\mathbb{R}) &\to D_n^{m-1}(\mathbb{R}) \\ f &\mapsto f' \end{aligned}$$

are continuous. In particular, $f_k \to f \Rightarrow f_k' \to f'$.

Differentiable functions

$D_n^m(\mathbb{R})$ — the space of n-ary m times differentiable functions:

Results: for all $m, n \in \mathbb{N}$,

1. $D_n^m(\mathbb{R})$ is a Hausdorff topological vector space;
2. The inclusion map $j\colon D_n^{m+1}(\mathbb{R}) \to D_n^m(\mathbb{R})$ is continuous;
3. There is an isomorphism $\gamma\colon C^m(\mathbb{R}^n) \leftrightarrow D_n^m(\mathbb{R})$, given by
$$\gamma\left(\tilde{f}\right)(x_1)(x_2)\ldots(x_n) = \tilde{f}(x_1, x_2, \ldots, x_n).$$
4. $D_n^m(\mathbb{R})$ is a Fréchet space.

References

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